

Smart India Hackathon (SIH 2026) – Grand Finale Pitch

PROJECT: KRUSHIRAKSHA | PROBLEM ID: 26131 | GOVT OF MAHARASHTRA

20-Minute Easy English Master Presentation Script · Team: Binary Beacons

Speaker & Role	Module & Core Topic (Simple English)	Time Allotted
Speaker 1: Problem Pitch Master (Team Lead)	Farmer pain points, story of Ramesh Patil, ₹1.4 Lakh Cr loss & KrushiRaksha vision	0:00 – 4:00 (4.0 Min)
Speaker 2: ML Engineer 1 (Vision & Live Camera)	Live YOLO Camera (38 FPS), Drone RTSP stream, Softmax Probability Bars & Confusion Matrix	4:00 – 7:30 (3.5 Min)
Speaker 3: ML Engineer 2 (Weather & Epidemiology)	Live Weather Telemetry, 48h early fungal alerts (Blitecast) & Safe Spray Window	7:30 – 10:45 (3.25 Min)
Speaker 4: ML Engineer 3 (Pest Traps & CIBRC Dosage)	IP102 Pink Bollworm trap counter, CIBRC organic + chemical dosage math & KVK alerts	10:45 – 13:45 (3.0 Min)
Speaker 5: Backend Engineer	FastAPI Backend, Outbreak GIS Heatmap (DBSCAN 10km Quarantine) & Offline 2G SMS / IVR	13:45 – 16:45 (3.0 Min)
Speaker 6: Frontend Developer	React 19 Fast UI, 11 Indian Languages translation, Audio Voiceout & 3 Role Portals	16:45 – 19:15 (2.5 Min)
Speaker 1: Closing & Summary	Summary, pilot impact (4 days to 14ms latency), Govt integration & Q&A; invitation	19:15 – 20:00 (45 Sec)

PART 1: THE PROBLEM STATEMENT PITCH (4.0 MINUTES)

Speaker 1: Team Lead & Problem Pitch Master | Time: 0:00 – 4:00

1. The Opening Story (0:00 - 1:15):

"Respected Judges and Mentors, A very good morning to all of you.

We are **Team Binary Beacons**, and we are presenting our solution for **Problem Statement ID 26131 — KrushiRaksha**.

Judges, imagine a farmer named Ramesh Patil from Nashik, Maharashtra. He invested all his life savings—around ₹1.5 Lakhs—into his 3-acre tomato and grape farm. Just two weeks before harvest, small watery dark spots appeared on the lower leaves.

Ramesh does not have an agricultural scientist nearby. Like 80% of Indian farmers, he plucked a leaf and rushed to the local pesticide shop. The shopkeeper gave him an expensive ₹2,500 chemical bottle without any scientific check. Ramesh sprayed it on his crop. But 4 hours later, heavy rain started, and the entire medicine washed away into the soil.

Three days later, the fungal disease Late Blight spread across his entire farm. His complete 3-acre crop was ruined. Total loss: 100%. Ramesh was left with heavy bank debt."

2. The Real Problem & Data in India (1:15 - 2:30):

"Judges, this is the story of millions of farmers across India today:

- **₹1.4 Lakh Crore Annual Loss:** According to government and ICAR reports, India loses 15% to 25% of its total crop yield every year purely due to pests and plant diseases.
- **The 4-Day Delay:** Traditional testing and officer visits take 3 to 7 days. But fungal diseases spread within 24 to 48 hours. By day 4, it is too late to save the crop.
- **Harmful Chemical Overdose:** Because farmers do not get exact dosage advice, they spray 35% excess chemical pesticides. This poisons the soil, pollutes groundwater, and leads to export rejections.
- **Shortage of Officers:** In India, there is only 1 extension officer for every 1,162 farmer families. Inspecting every single farm physically is impossible."

3. Why Existing Mobile Apps Fail & The KrushiRaksha Solution (2:30 - 4:00):

"There are many mobile apps on the market, but why did they fail? Because:

1. They only take a single photo, instead of live camera scanning and drone video feeds.
2. They ignore the weather—fungus only grows when high humidity and leaf wetness are present.
3. They work like commercial pesticide shops, pushing chemicals instead of safe organic methods and exact CIBRC dosage.

To solve this, we built **KrushiRaksha**—an AI-powered platform that connects Farmers, Extension Officers, and the Government on one single system. Let us invite our Lead ML Engineer to show you our Live AI Camera."

PART 2: LIVE AI CAMERA, PREDICTION PROBABILITIES & CONFUSION MATRIX (3.5 MIN)

Speaker 2: ML Lead 1 (Vision & Live Camera) | Time: 4:00 – 7:30

Live Demo & Technical Walkthrough:

"Thank you. On the live screen, Pillar 1 is our **AI Diagnosis & Live YOLO Camera**:

- **Live YOLOv8 Camera Vision:** The farmer does not need to click 50 separate photos. They just turn on the live camera. Our model runs at **38 Frames Per Second with a super-fast 14ms latency**. It draws live bounding boxes on the leaf, detecting active disease spots and yellowing in real time. It supports mobile lenses, tractor cameras, and drone video feeds.
- **Photo Upload with Cloud Vision API:** When a farmer uploads a leaf photo, our vision model (Google Gemini 1.5 Flash / ResNet-18) reads the image and renders a **Grad-CAM heatmap** showing the exact infected area on the leaf.
- **Prediction Probabilities (Multi-Class Softmax Bars):** Our model does not just give a single name. It displays the full probability breakdown across all candidate classes—for example, Mango Anthracnose: 87.4%, Healthy Leaf: 7.8%, Powdery Mildew: 3.1%. This proves to evaluators that the model evaluates all possibilities scientifically.
- **Model Validation Confusion Matrix:** By clicking the Confusion Matrix button, judges can see our benchmark results across 54,000+ images with **96.8% Accuracy and a 0.962 F1-Score**.

Now, photos only show disease after it appears. But weather causes the disease. Let our 2nd ML Engineer explain our live weather engine."

PART 3: LIVE WEATHER RISK & SAFE SPRAY WINDOW (3.25 MINUTES)

Speaker 3: ML Lead 2 (Epidemiology & Weather AI) | Time: 7:30 – 10:45

Live Demo & Technical Walkthrough:

"Thank you. Pillar 2 is our **Live Weather Risk Forecaster**:

- **Live Weather Telemetry:** We connected the Open-Meteo API to fetch live weather data for any Indian farm, along with **1-Click GPS Auto-Location**. It tracks Temperature, Humidity, Leaf Wetness, and Wind Speed in real time.
- **48-Hour Early Disease Warning (Blitecast & 3-10 Rule):** When humidity crosses 85% and leaves stay wet for over 10 hours, our algorithm gives an **early alert 48 to 72 hours in advance**, warning the farmer that fungal sporulation is about to start.
- **Safe Spray Window (When to Spray Medicines):** Farmers lose thousands of rupees when rain washes away their sprayed chemicals. Our system checks the 6-hour rain forecast and wind speed to clearly display **'SAFE SPRAY WINDOW ACTIVE'** or **'UNSAFE SPRAY WINDOW'** to protect the farmer's investment.

Now, our 3rd ML Engineer will show how we calculate the exact medicine dosage safely."

PART 4: PEST TRAP COUNTING & CIBRC SAFE MEDICINE DOSAGE (3.0 MINUTES)

Speaker 4: ML Lead 3 (Pest Traps & CIBRC Dosage) | Time: 10:45 – 13:45

Live Demo & Technical Walkthrough:

"Thank you. Pillar 3 is our **CIBRC-Approved Medicine Dosage & IPM Calculator**:

- **3-Step Safe Solution:** Whenever disease is detected, we provide 3 hierarchical steps:
 1. Step 1 (Cultural Farming Methods): Removing infected leaves and maintaining row spacing.
 2. Step 2 (Biological Control): Organic bio-agents like *Trichoderma harzianum* or *Pseudomonas*.
 3. Step 3 (CIBRC Certified Chemicals): Government-approved chemical molecules (like Copper Oxychloride 50% or Mancozeb) without any commercial brand bias.
 - **Tank Dosage Calculator:** The farmer inputs their land acreage and 15-Liter knapsack tank size. The calculator gives the exact grams of medicine needed and the **Pre-Harvest Interval (PHI)** safety days so no toxic chemical residue remains in the harvest.
 - **Pest Trap Counter (Pink Bollworm):** In cotton fields, our IP102 AI model counts moths caught in pheromone traps. When catches exceed **8 moths per trap per night**, the system automatically creates an urgent ticket for the nearest KVK agricultural expert.
- Now, our Backend Engineer will explain how the server and offline features work."

PART 5: FASTAPI BACKEND, OUTBREAK GIS & OFFLINE 2G SMS (3.0 MINUTES)

Speaker 5: Backend Engineer | Time: 13:45 – 16:45

Live Demo & Technical Walkthrough:

"Thank you. Pillars 4 and 5 are our backend architecture:

- **High-Speed FastAPI Backend:** Our backend is written in Python FastAPI, delivering sub-50ms API response times for camera streams, image analysis, and spatial queries.
- **DBSCAN Outbreak GIS Clustering:** When multiple farmers upload disease scans in a district, our spatial algorithm clusters their GPS locations. If an infection grows within a 10km radius, it automatically draws a **10km Quarantine Containment Zone** on the Government map to stop the outbreak early.
- **Offline 2G SMS / IVR Gateway (*99*26131# & 1800-KRUSHI):** For the 40% of smallholder farmers who only have a simple keypad button phone with zero internet, they can send a simple text message: `KRUSHI TOMATO SPOTS NASHIK`. Our server replies within 2 seconds via normal SMS with the exact medicine dosage in their local language.

Now, our Frontend Developer will walk you through the user interface and 11 language translations."

PART 6: REACT 19 FRONTEND, 11 INDIAN LANGUAGES & 3 ROLES (2.5 MINUTES)

Speaker 6: Frontend Developer | Time: 16:45 – 19:15

Live Demo & Technical Walkthrough:

"Thank you. For farmers, technology is only useful if it speaks their mother tongue:

- **Full 11-Language Translation:** As you can see on screen, when we switch to Marathi, Hindi, Punjabi, Kannada, Telugu, Tamil, Bengali, Gujarati, Malayalam, or Odia—the entire website, buttons, camera HUD, weather cards, and medicine prescriptions translate instantly without errors.
- **Voiceout Audio Advisory:** Clicking 'Listen Advisory' plays loud, clear speech in the farmer's native dialect for hands-free field use.
- **3 Dedicated Role Hubs:**
 1. ■ **Farmer Hub:** Simple mobile buttons for leaf scanning, weather alerts, and dosage calculation.
 2. ■■■ **Officer Hub:** Field visit verification queue and KVK ticket resolution.
 3. ■■ **Government Command Center:** State-wide epidemic maps, medicine supply tracking, and 1-click broadcast alerts to 50,000 farmers.

Let me invite our Team Lead for the final wrap-up."

PART 7: GRAND SUMMARY & FINAL WRAP-UP (45 SECONDS)

Speaker 1: Team Lead | Time: 19:15 – 20:00

"Judges, to conclude:

KrushiRaksha is not a concept—it is a **100% functional, live deployed platform** right now on Vercel (krushiraksha.vercel.app) and GitHub Pages.

In our simulated trials, disease detection time dropped from **4 days to 14 milliseconds**, and farmer pesticide costs dropped by **28%**. We are ready for direct integration with the Government of Maharashtra's **MahaAgri** portal and the national **e-NAM** platform.

Thank you so much, Judges! We are now excited to take your questions!"

JURY CROSS-EXAMINATION DEFENSE (EASY ENGLISH Q&A; CHEAT SHEET)

#	Expected Hard Question	Who Answers	What to Say (Simple, Winning English Response)
Q1	How is your model different from other open-source models?	ML Lead 1 (Speaker 2)	Sir, standard models only work on clean lab photos. We trained on PlantVillage & IP102 with real sunlight, shadows, camera blur, and drone angles. Plus, we combine image pixels with Grad-CAM heatmaps and live weather rules.
Q2	What if a farmer cannot read or does not have a smartphone?	Backend Lead (Speaker 5)	Sir, we have two accessibility solutions: 1. A 2G SMS and IVR Toll-Free service (*99*26131#) for simple keypad phones. 2. Native voice synthesis in 11 Indian languages that speaks the medicine instructions aloud.

Q3	How do you prevent pesticide companies from promoting their brand?	ML Lead 3 (Speaker 4)	Sir, KrushiRaksha is strictly generic molecule-first. We only recommend government-approved CIBRC active ingredients (like Copper Oxychloride 50%). Brand names are shown only as non-sponsored reference examples.
Q4	What is your business model and sustainability plan?	Team Lead (Speaker 1)	Sir: 1. B2G SaaS subscriptions from State Agriculture Departments for Outbreak GIS. 2. B2B Edge API licensing for drone spraying cooperatives. 3. 100% Free Forever for farmers as digital public infrastructure.